

Math 141, S10

Name: Answer

Quiz 6

Section:

SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Find the derivatives of the following functions.

(a) $f(t) = t^3 \cos t$

$$\begin{aligned} f'(t) &= t^3 (\cos t)' + \cos t (t^3)' \\ &= -t^3 \sin t + 3t^2 \cos t \end{aligned}$$

(b) $g(x) = \left(\frac{x^2 + 1}{x^2 - 1} \right)^6$

$$\begin{aligned} g'(x) &= 6 \left(\frac{x^2 + 1}{x^2 - 1} \right)^5 \frac{d}{dx} \left[\frac{x^2 + 1}{x^2 - 1} \right] \\ &= 6 \left(\frac{x^2 + 1}{x^2 - 1} \right)^5 \frac{(x^2 - 1) \frac{d}{dx} [x^2 + 1] - (x^2 + 1) \frac{d}{dx} [x^2 - 1]}{(x^2 - 1)^2} \\ &= 6 \left(\frac{x^2 + 1}{x^2 - 1} \right)^5 \frac{(2x^3 - 2x) - (2x^3 + 2x)}{(x^2 - 1)^2} \\ &= 6 \left(\frac{x^2 + 1}{x^2 - 1} \right)^5 \frac{(-4x)}{(x^2 - 1)^2} \rightarrow \text{ok stop here.} \\ &= -24 \frac{x(x^2 + 1)^5}{(x^2 - 1)^7} \end{aligned}$$

(c) $h(x) = e^{\sin x}$

$$\begin{aligned} h'(x) &= e^{\sin x} \frac{d}{dx} [\sin x] \\ &= e^{\sin x} \cos x \end{aligned}$$